

Thin metallic strips can support long-range surface plasmons—a particular type of surface plasmon mode characterised by electromagnetic fields mostly contained in the region outside of the metal, i.e., in dielectric medium. Jung *et al* in 2007 experimentally confirmed that long-range surface plasmons could transfer data signal as well as the carrier light. In a demonstration, a 10Gbps signal was transmitted over a thin metallic strip (14nm thick, 2.5µm wide and 4cm long gold strip).

Furthermore, to reduce the propa-

One primary goal of this field is to develop new optical components and systems that are of the same size as today's smallest integrated circuits and that could ultimately be integrated with electronics on the same chip. The next step will be to integrate the components with an electronic chip to demonstrate plasmonic data generation, transport and detection.

Plasmon waves on metals behave much like light waves in glass. That means engineers can use techniques like multiplexing or sending multiple

tion? Surface plasmon polaritons are usually excited using far-field optical techniques, which have a higher resolution than plasmonic phenomena under investigation. However, for true nanoscale plasmonic studies, a surface-plasmon-polariton point source with nanoscale dimensions is required.

"What are the fundamental processes that determine the losses of surface plasmon polaritons?" is another important question. Practically, plasmon experiments are performed on poly-crystalline surfaces, and the limits to the losses due to surface roughness, grain boundaries, etc are not known.

Surface plasmons propagate along the chain of nanoparticles, but the losses are high. On the other hand, propagation losses are low in the case of nanowires, which leaves open the possibility of surface-plasmon optical devices.

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To manipulate surface plasmon polaritons on a surface, reflectors are needed. So far, macroscopic Bragg reflectors structured into the surface have been used. For true nanoscale integration, nanoscale surface plasmon polariton mirrors are required. Once these are realised, nanoscale cavities to confine surface plasmon polaritons can also be designed. The limits to the mode volume and quality factor of plasmonic cavities are not yet known.

Finally, the use of a particle beam rather than a light beam to excite surface plasmon polaritons raises questions and novel opportunities regarding the selectivity with which surface plasmon modes with different symmetry can be excited. ●

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gation loss, Jin Tae Kim *et al* fabricated a low-loss, long-range surface plasmon polariton waveguide in an ultraviolet-curable acrylate polymer having low refractive index and absorption loss. A 14nm thick and 3µm wide metallic strip clad in acrylate polymer material shows a loss of 1.72 dB/cm.

Rashid Zia *et al* obtained the numerical solution by using the full-vectorial magnetic field finite-difference method for 55nm thick and 3.5µm wide strip on glass at a wavelength of 800 nm and noted that surface plasmons are supported on both sides of the strip and can propagate independently.

Alexandra *et al* in year 2008 suggested that triangular metal wedge could guide surface plasmons at telecommunication wavelength. It was experimentally observed that 1.43-1.52µm wavelength can propagate over a distance of about 120 µm with confined-mode width of 1.3 µm along a 6µm high and 70.5° angled triangular gold wedge.

Future directions

In the field of plasmonics, studying the way light interacts with metallic nanostructures will make it easier to design new optical material devices.

waves.

Plasmon sources, detectors and wires as well as splitters and even plasmonsters can be developed. Applications mainly depend on controlling the losses and the cost of nanofabrication techniques. Enhanced and directed emission of semiconductor luminescence (quantum dots) may well find commercial application in plasmon-assisted lighting in the near future.

Finally, plasmonic nanocircuits combine a high bandwidth with a high level of compaction and make plasmonic components promising for all-optical circuits. Plasmonic wires will act as high-bandwidth free-waves across the busiest areas of the chip. Plasmons can ferry data along computer chips. Plasmonic switches required for this are under development. Rotaxanes molecule is being used for the purpose. Change in the shape of the molecule is the principle of this molecular switch.

Challenges remaining

Despite many advances in the field of plasmonics, several important open questions and problems remain. For example, how can plasmons be efficiently excited with nanoscale resolu-

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